MLOPS Assignment 2

Name: Nidhi
USN: 4GW23CI035

1. Which run had the lowest latency, and why might that be?

- run_batch8 had the lowest latency, at 0.120 s.
- run_batch4 took 0.133 s, and run_batch1 was the slowest at 0.226 s.
- With batch size 8, the model processes 8 sentences together in one forward pass, so the 10 sentences need only 2 model calls.
- Fewer calls means less overhead per call, so the total time is lower.
- run_batch1 made 10 separate calls, and since it ran first it also paid a one-time warm-up cost.
- So latency decreases as batch size increases.

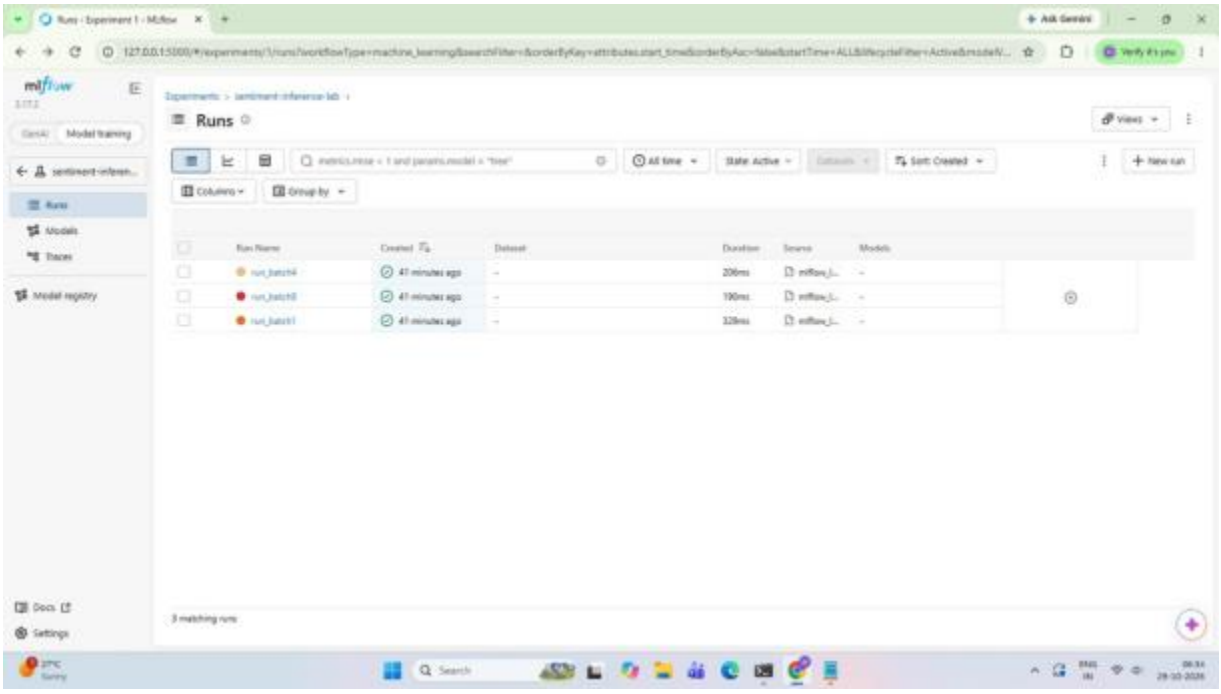
2. Did accuracy change between runs? Why or why not?

- No, accuracy stayed at 1.0 (100%) in all three runs.
  Every run used the same pre-trained DistilBERT model with the same weights.
- All runs were tested on the same 10 sentences.
- Inference is deterministic, meaning the same input always gives the same output.
- Batch size only changes how fast the sentences are processed, not the predictions.
- Average confidence was also the same (0.99965) in all runs.

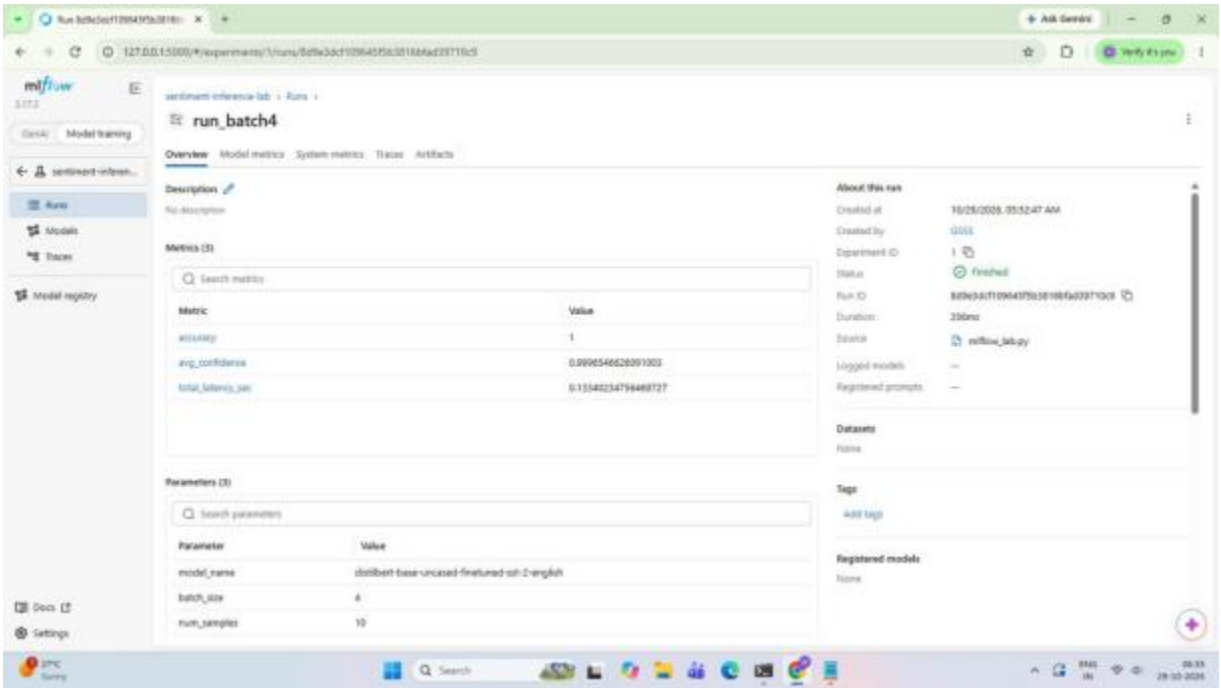
3. Name two benefits of using MLflow here instead of printing results in the terminal.

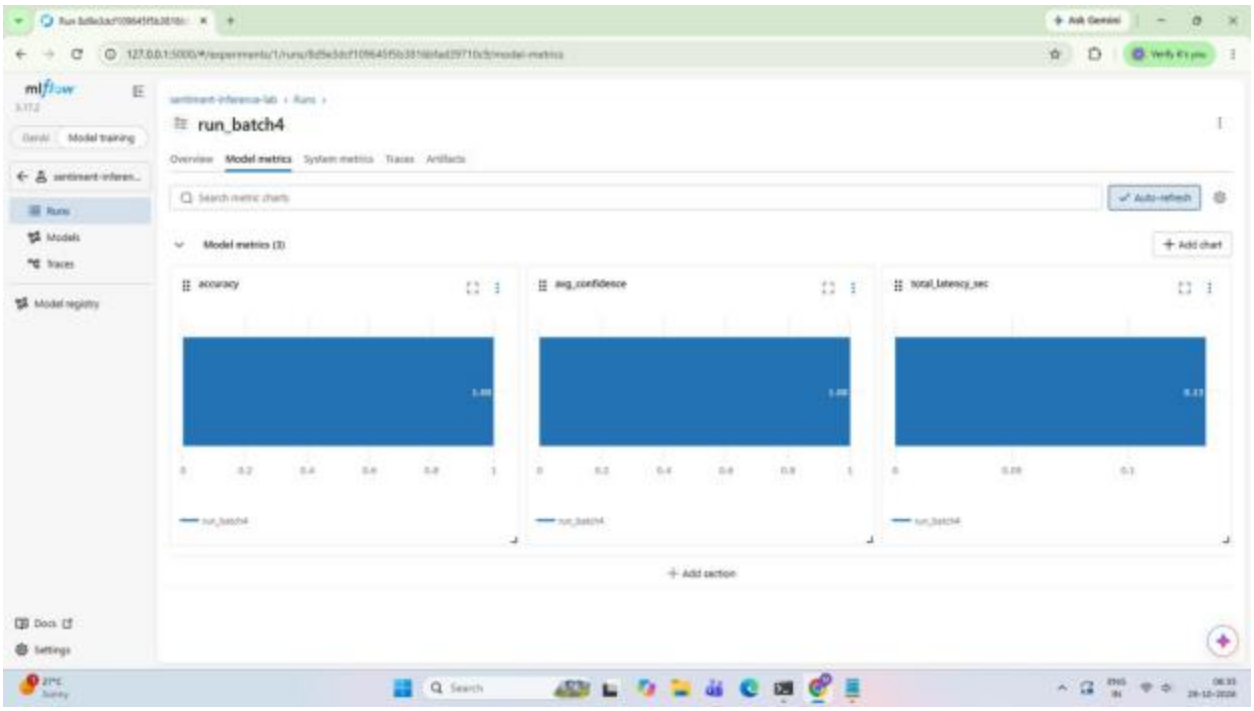
- Reproducibility and record-keeping: the parameters, metrics and artifacts (predictions.csv) of every run are saved permanently. Terminal output is lost once the window closes.
- Easy comparison: runs can be viewed and compared side by side in the MLflow UI, with tables and charts, instead of reading through printed lines.

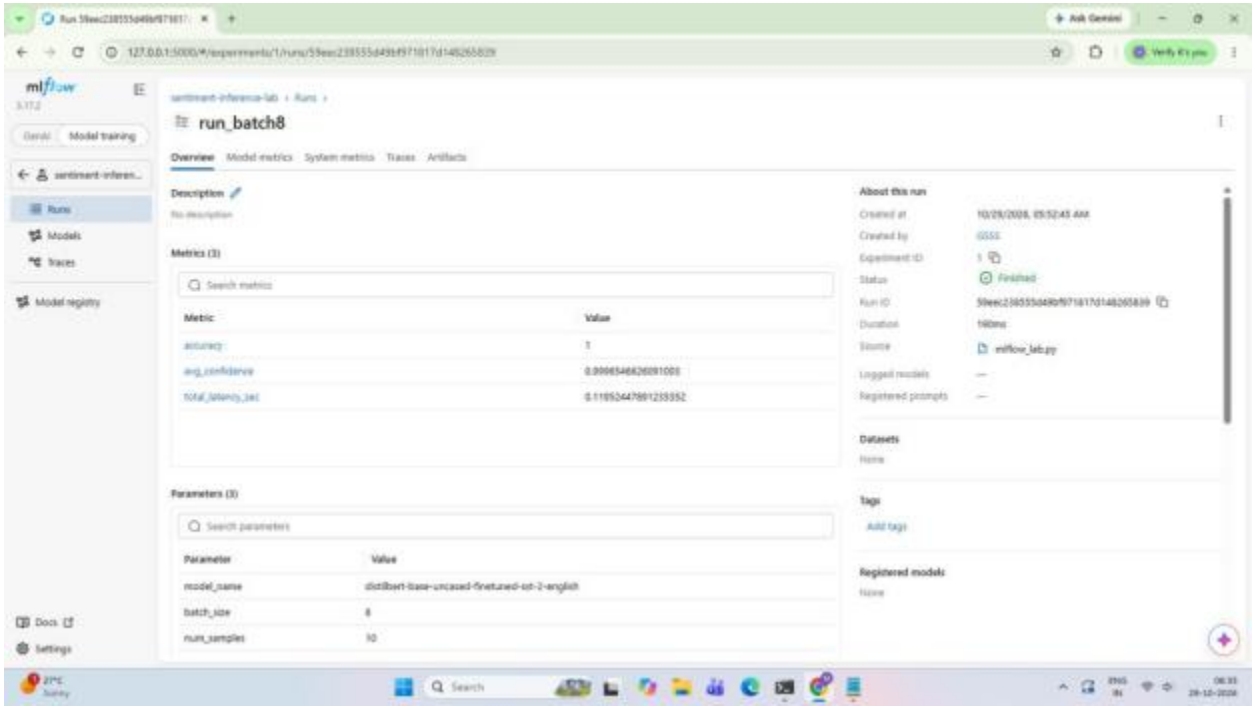
Screenshot of the MLflow runs table:

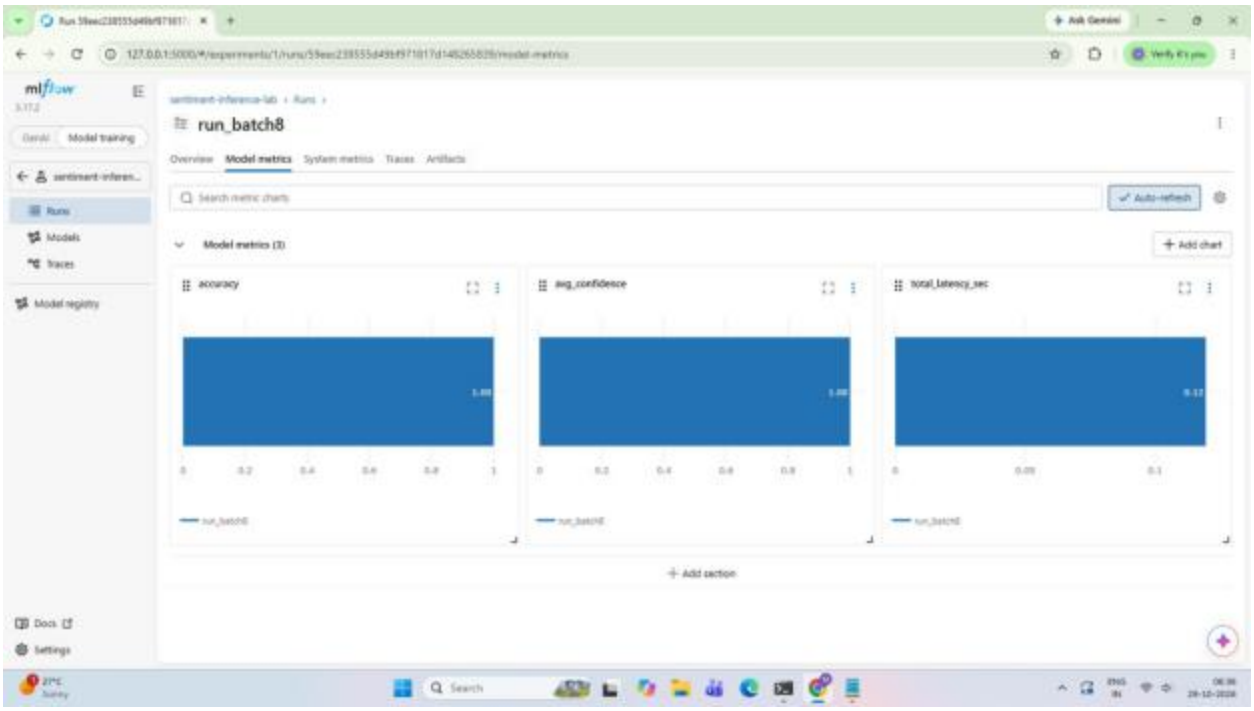


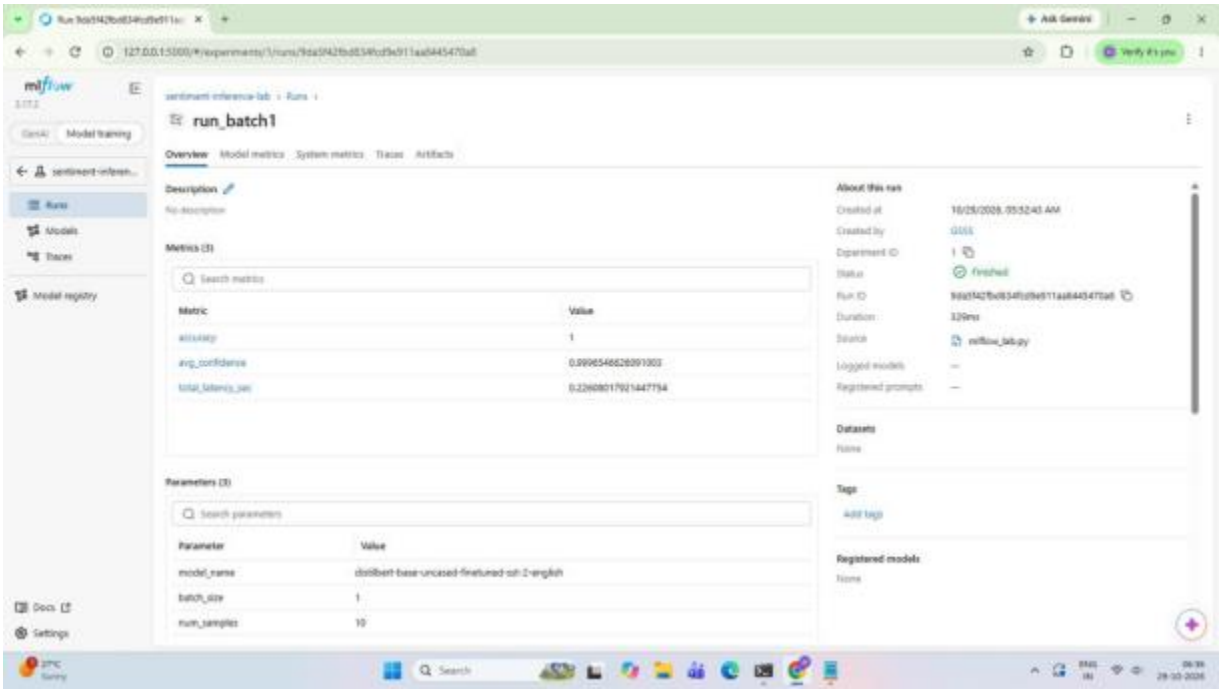
ML FLOW FULL SCREEN

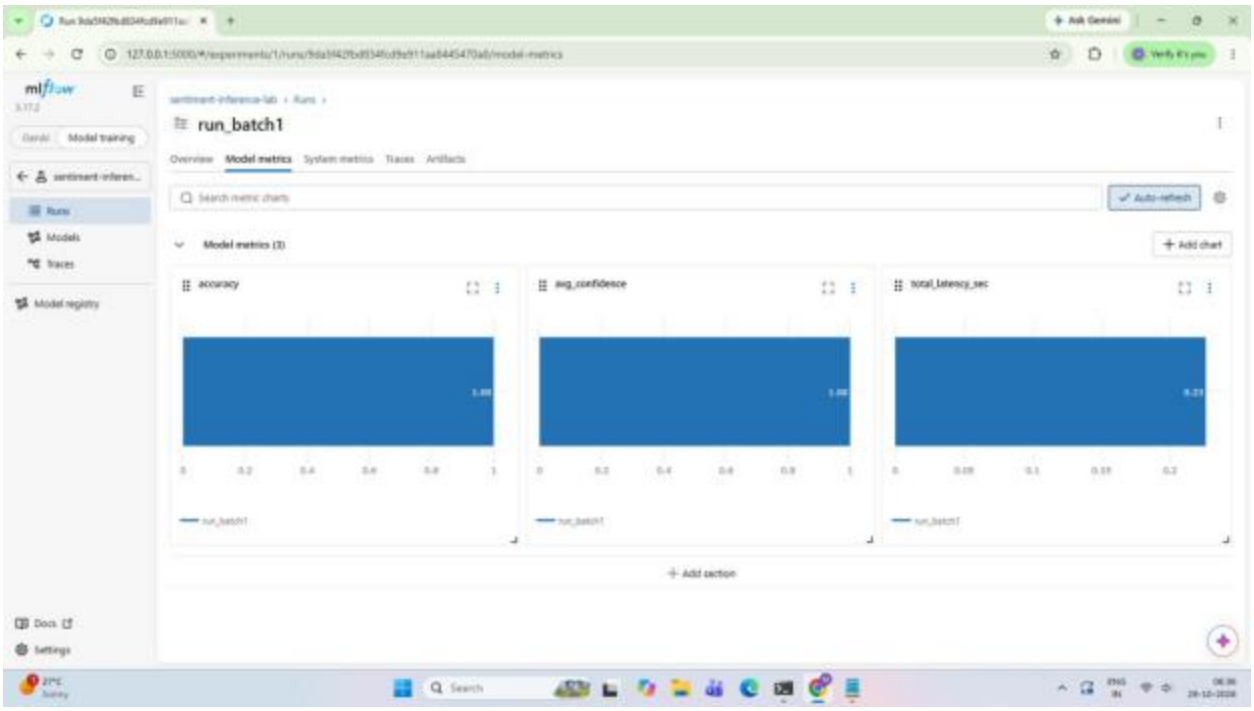



mlflow
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Model metrics (3)
accuracy
avg_confidence
total_latency_sec
Add section

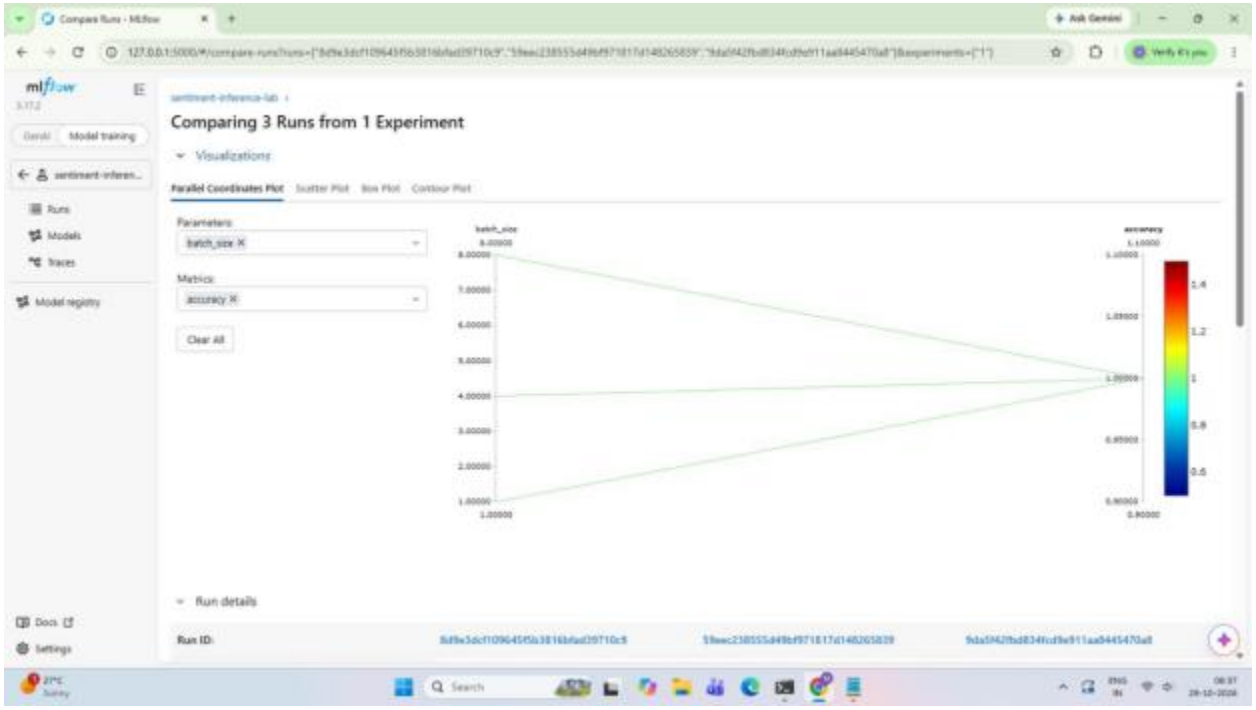

mlflow
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
Metrics (3)
Metric
Value
accuracy
avg_confidence
total_latency_sec
Parameters (3)
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
num_samples
About this run
Datasets
Tags
Registered models


mlflow
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Model metrics (3)
+ Add chart
accuracy
avg_confidence
total_latency_sec
+ Add section
Runs
Models
Traces
Model registry
Docs
Settings

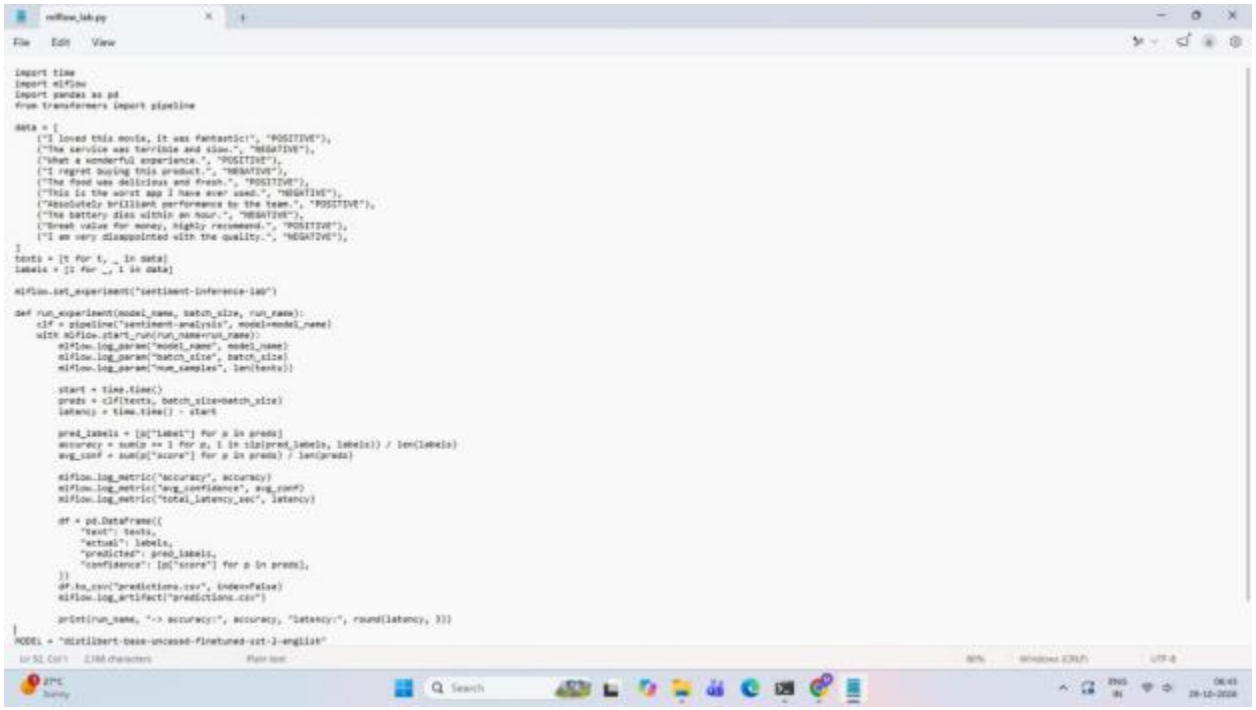

mlflow
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
Metrics (3)
Metric
Value
accuracy
1
avg_confidence
0.9996546826091303
total_latency_sec
0.22608017921447754
Parameters (3)
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
1
num_samples
10
About this run
Created at
10/25/2025, 05:52:43 AM
Created by
GSS
Experiment ID
1
Status
Finished
Run ID
Duration
329ms
Source
mlflow_lab.py
Logged models
Registered prompts
Datasets
None
Tags
Add tags
Registered models
None
Runs
Models
Traces
Model registry
Docs
Settings



Screenshot of the Compare view:



mlflow_lab.py file:



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

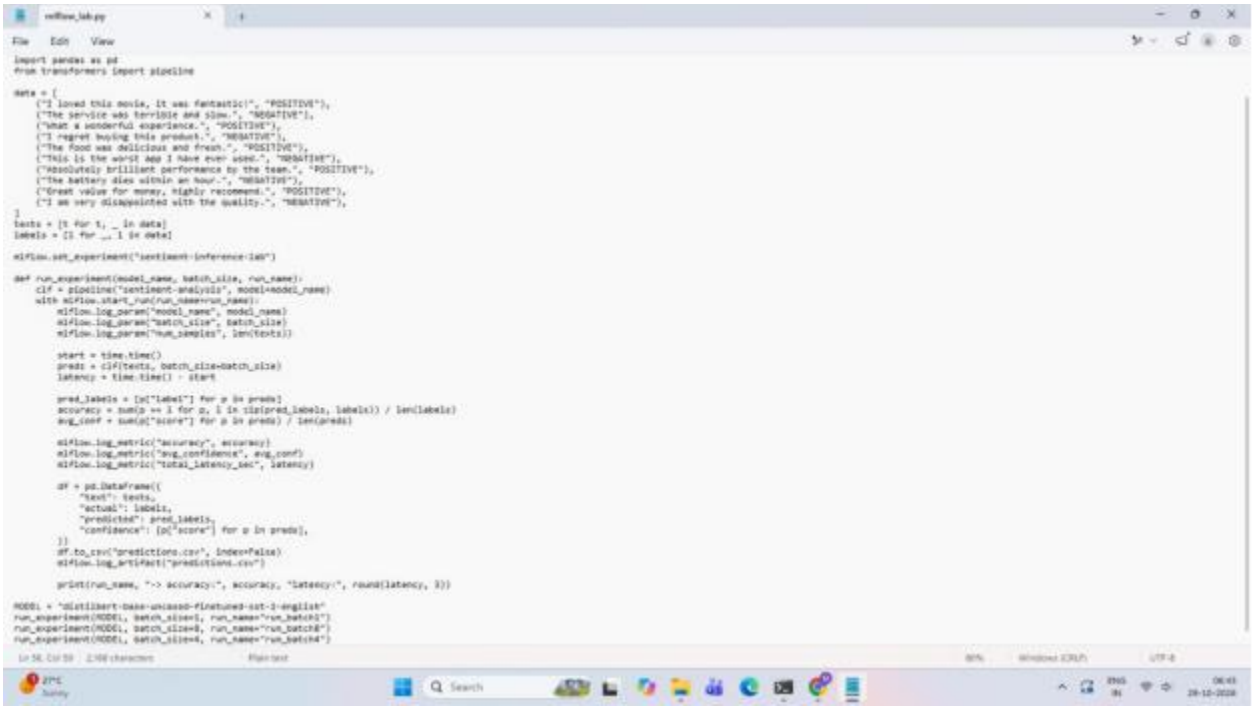
        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
```



```
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

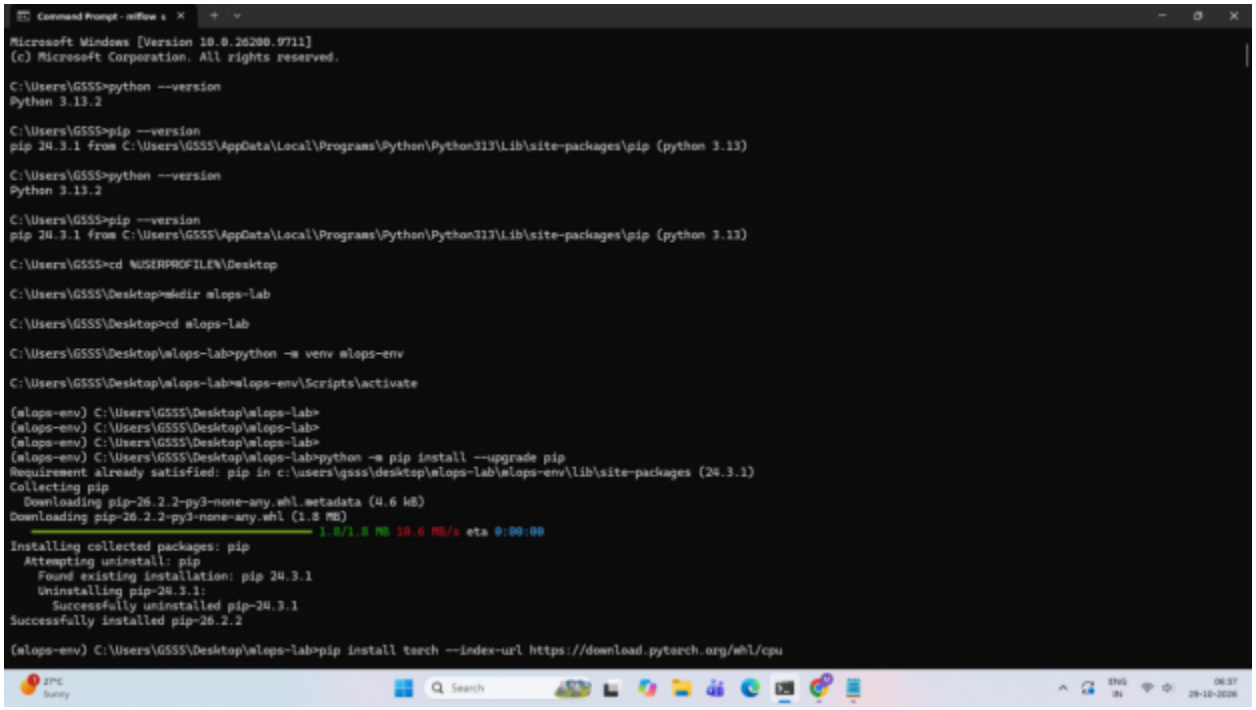
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

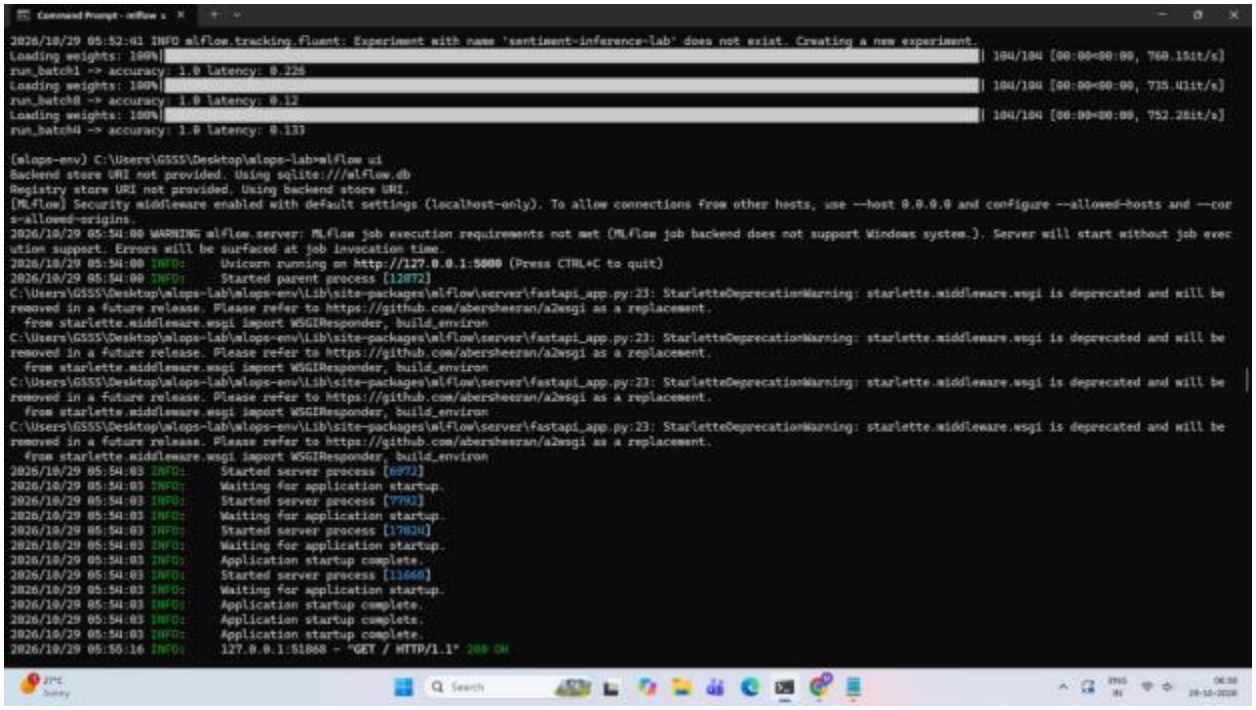
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

Execution:


Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>cd %USERPROFILE%\Desktop
C:\Users\GSSS\Desktop>mkdir mlops-lab
C:\Users\GSSS\Desktop>cd mlops-lab
C:\Users\GSSS\Desktop\mlops-lab>python -m venv mlops-env
C:\Users\GSSS\Desktop\mlops-lab>mlops-env\Scripts\activate
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>python -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\gsss\desktop\mlops-lab\mlops-env\lib\site-packages (24.3.1)
Collecting pip
Downloading pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 10.6 MB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 24.3.1
Uninstalling pip-24.3.1:
Successfully uninstalled pip-24.3.1
Successfully installed pip-26.2.2
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>pip install torch --index-url https://download.pytorch.org/whl/cpu


2026/10/29 05:52:41 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%| 104/104 [00:00<00:00, 760.15it/s]
run_batch1 -> accuracy: 1.0 latency: 0.226
Loading weights: 100%| 104/104 [00:00<00:00, 735.41it/s]
run_batch0 -> accuracy: 1.0 latency: 0.12
Loading weights: 100%| 104/104 [00:00<00:00, 752.28it/s]
run_batch4 -> accuracy: 1.0 latency: 0.133
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 05:54:00 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/29 05:54:00 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 05:54:00 INFO: Started parent process [12872]
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/29 05:54:03 INFO: Started server process [6972]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [7792]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [17824]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Started server process [11660]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:55:16 INFO: 127.0.0.1:51868 - "GET / HTTP/1.1" 200 OK

1. **Which run had the lowest latency, and why might that be?**

   run_batch8 had the lowest latency (0.120 s), followed by run_batch4 (0.133 s) and run_batch1 (0.226 s). With batch size 8, the model processes several sentences in one forward pass, so it makes fewer calls and has less overhead per call. run_batch1 makes 10 separate calls, and since it ran first it also paid a one-time warm-up cost

2. **Did accuracy change between runs? Why or why not?**

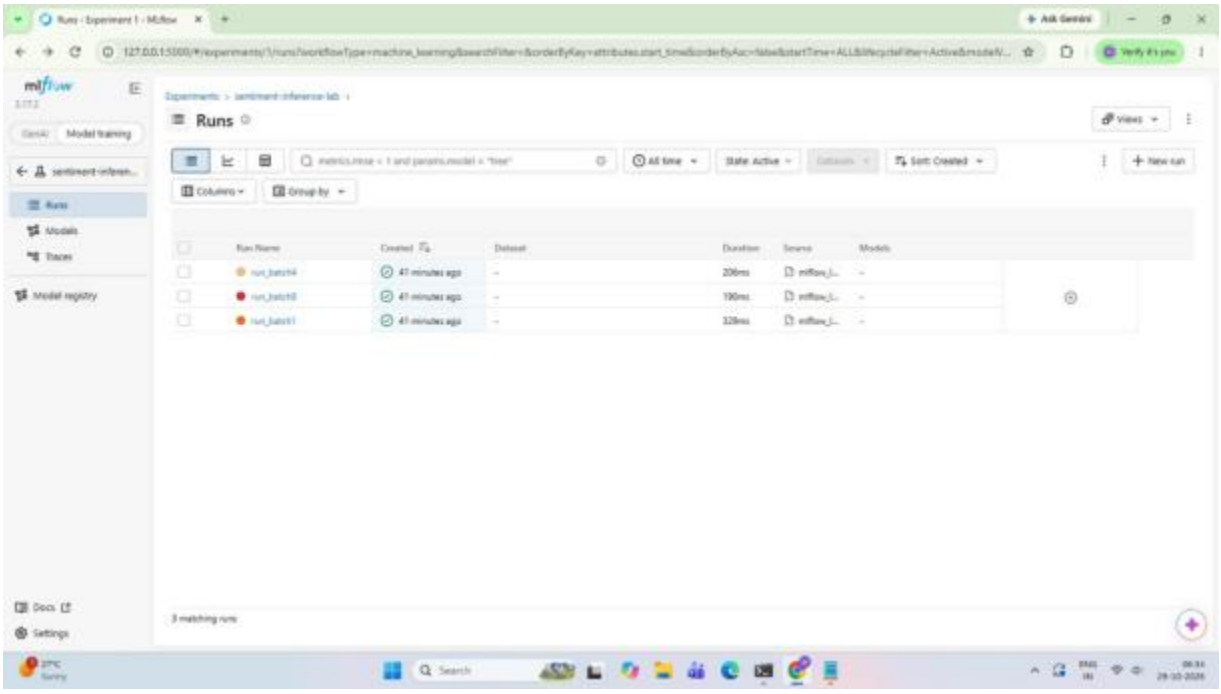
   No, accuracy stayed at 1.0 (100%) in all three runs. Every run used the same pre-trained model, with the same weights, on the same 10 sentences, and inference is deterministic. Batch size only changes how fast the sentences are processed, not the predictions

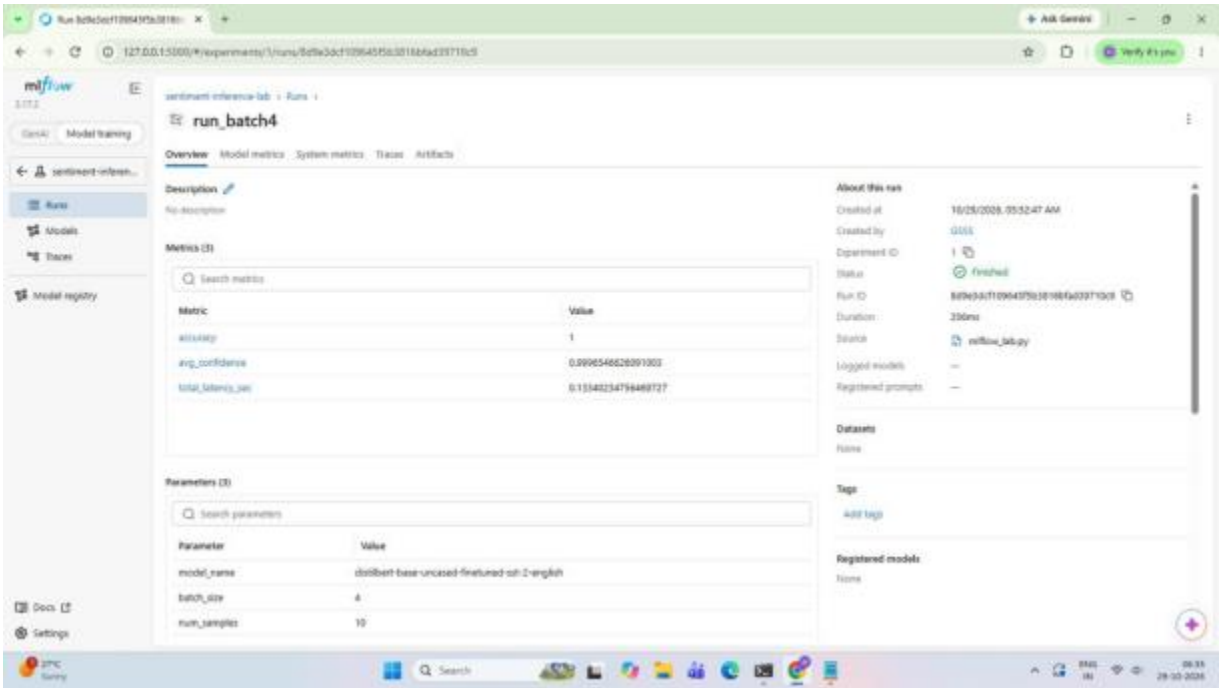
3. **Name two benefits of using MLflow here instead of printing results in the terminal.**

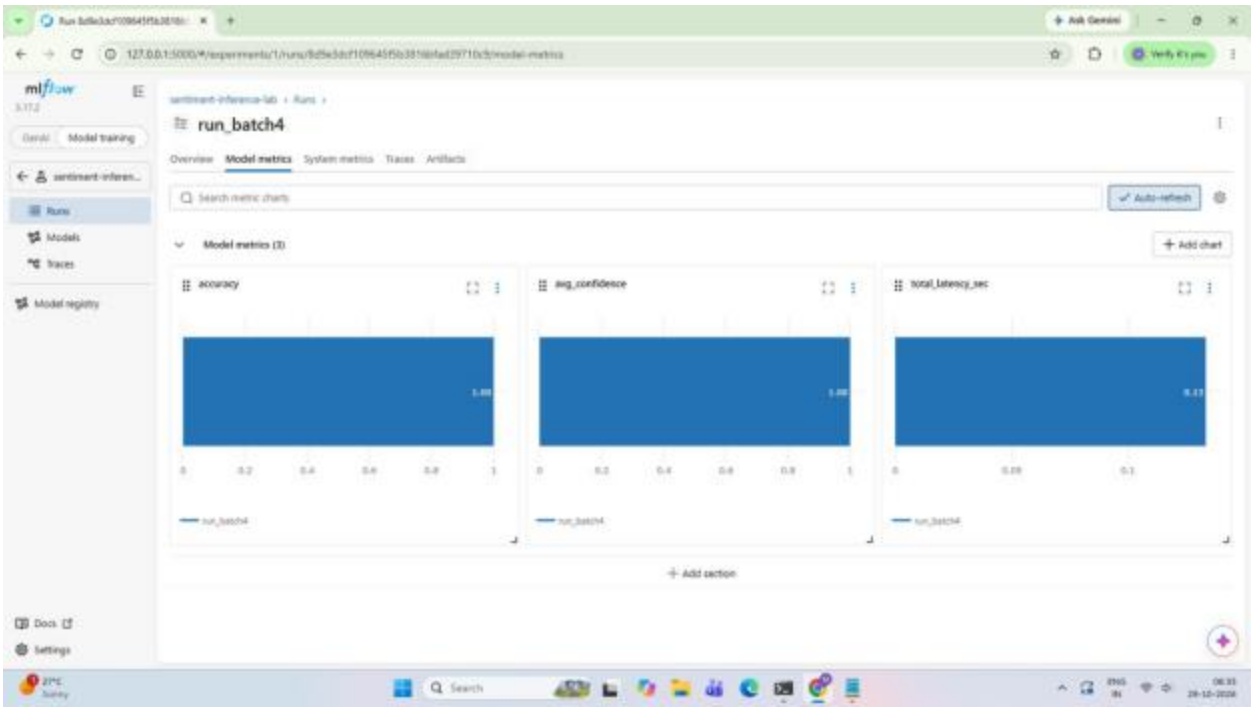
   Two benefits of MLflow:

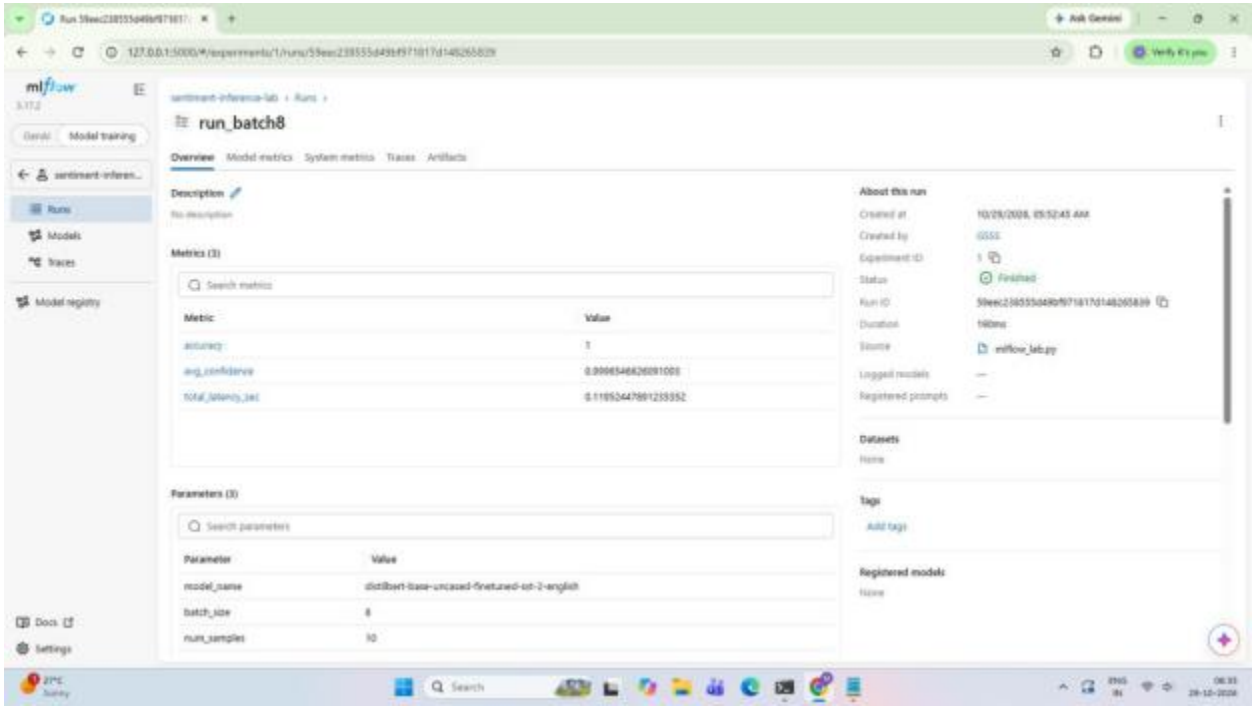
   - Reproducibility and record-keeping: the parameters, metrics and artifacts (predictions.csv) of every run are saved permanently. Terminal output is lost once the window closes.
   - Easy comparison: you can view and compare runs side by side in the UI, with tables and charts, instead of reading through printed lines

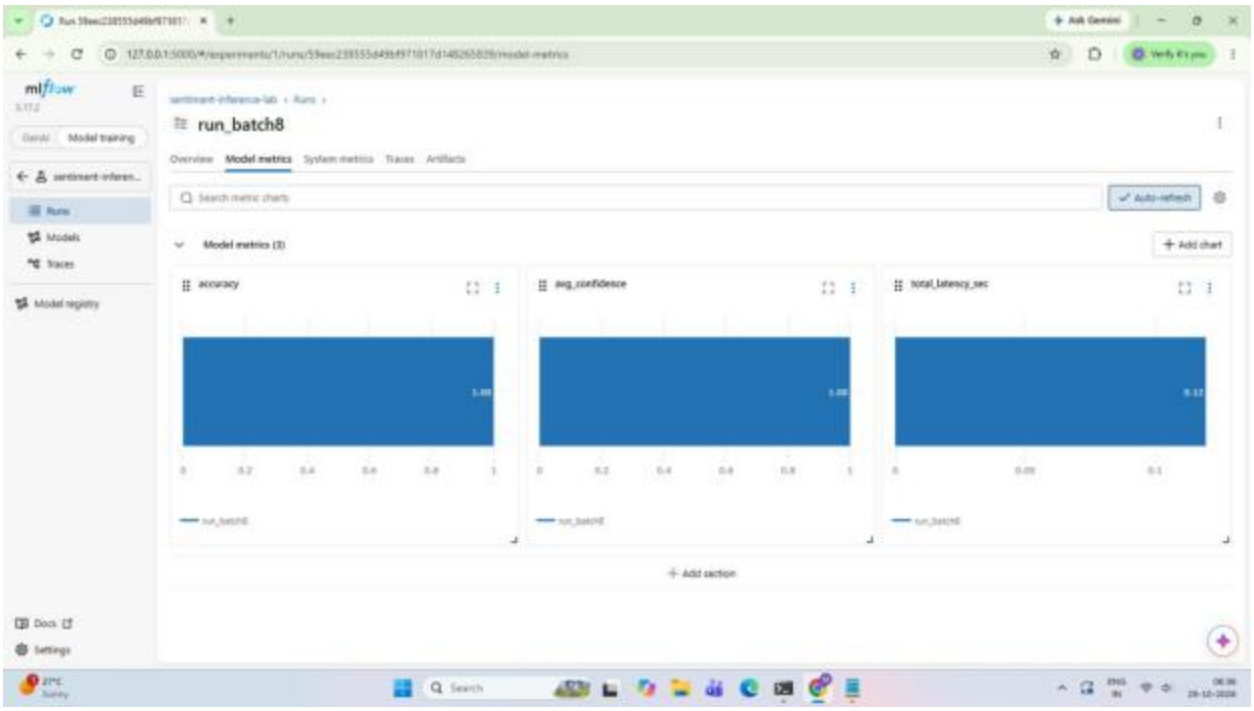
Screenshot of the MLflow runs table:

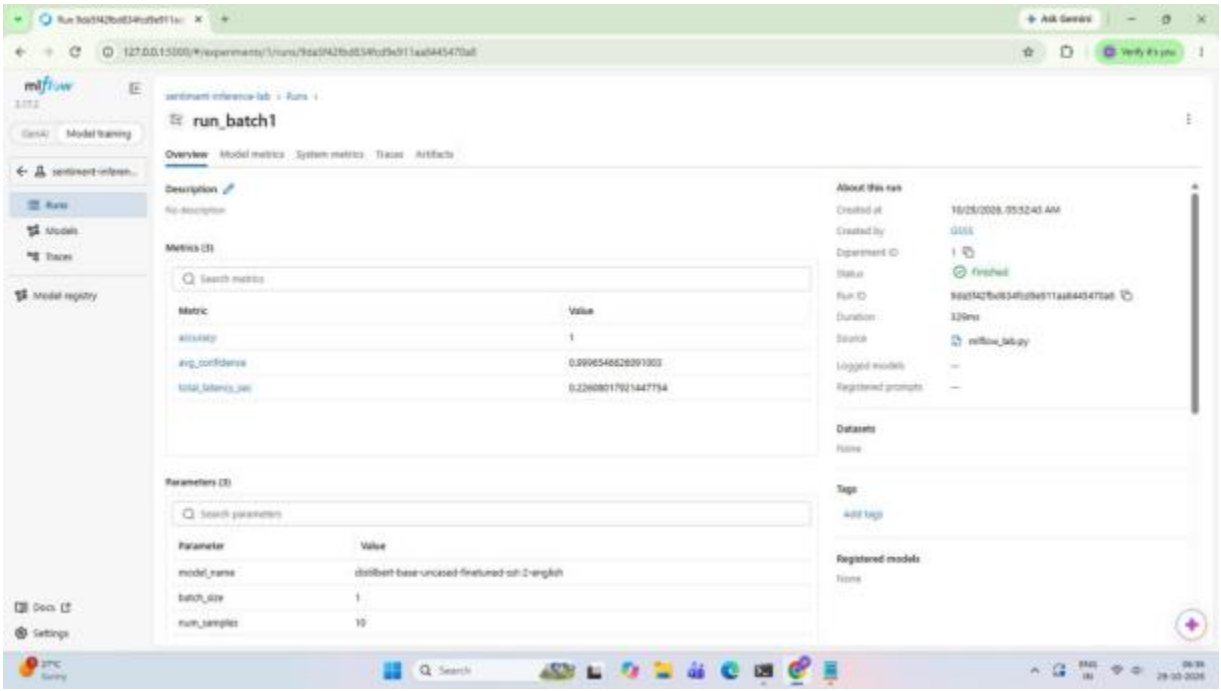


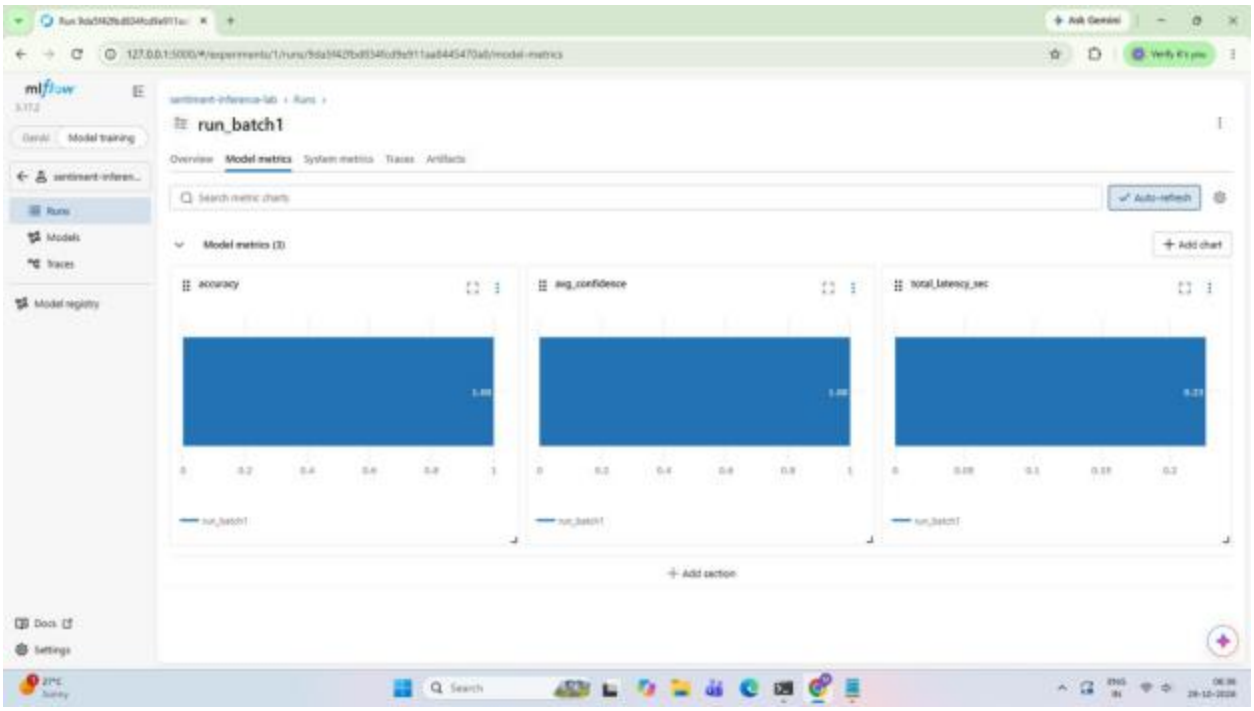



mlflow
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Model metrics (3)
accuracy
avg_confidence
total_latency_sec
Add section

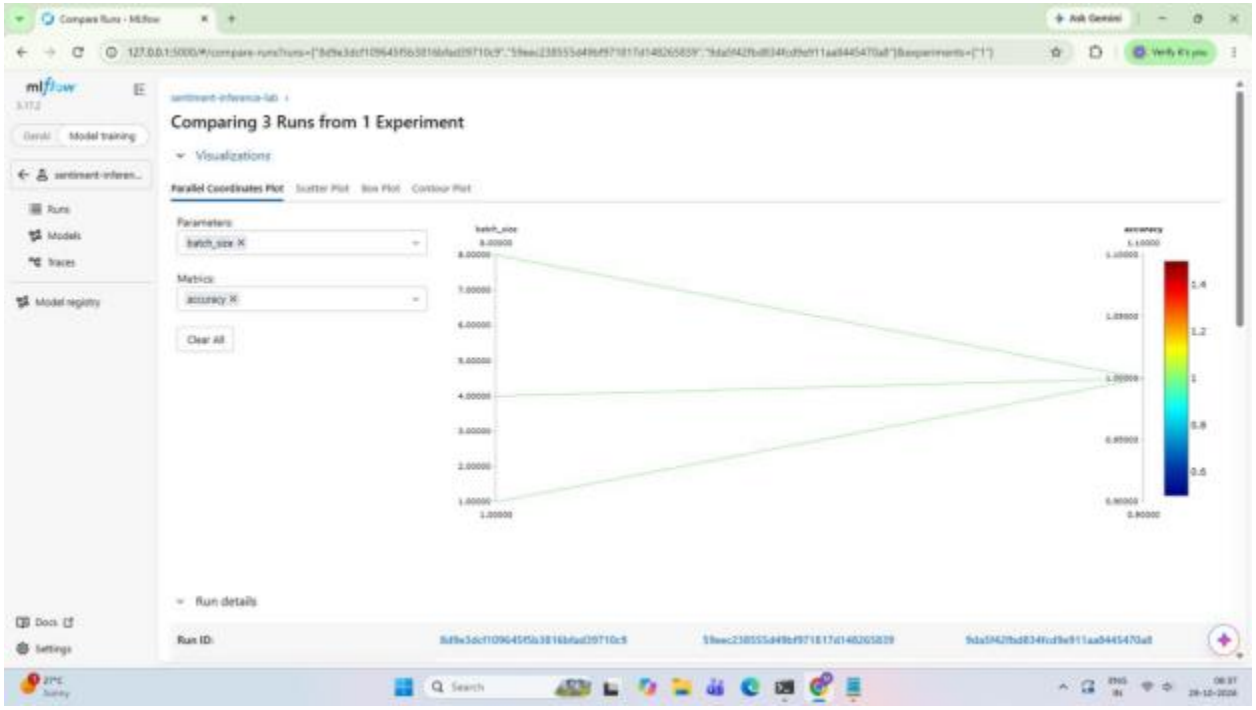

mlflow
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
Metrics (3)
Metric
Value
accuracy
avg_confidence
total_latency_sec
Parameters (3)
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
num_samples
About this run
Datasets
Tags
Registered models


mlflow
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Model metrics (3)
+ Add chart
accuracy
avg_confidence
total_latency_sec
+ Add section
Runs
Models
Traces
Model registry
Docs
Settings

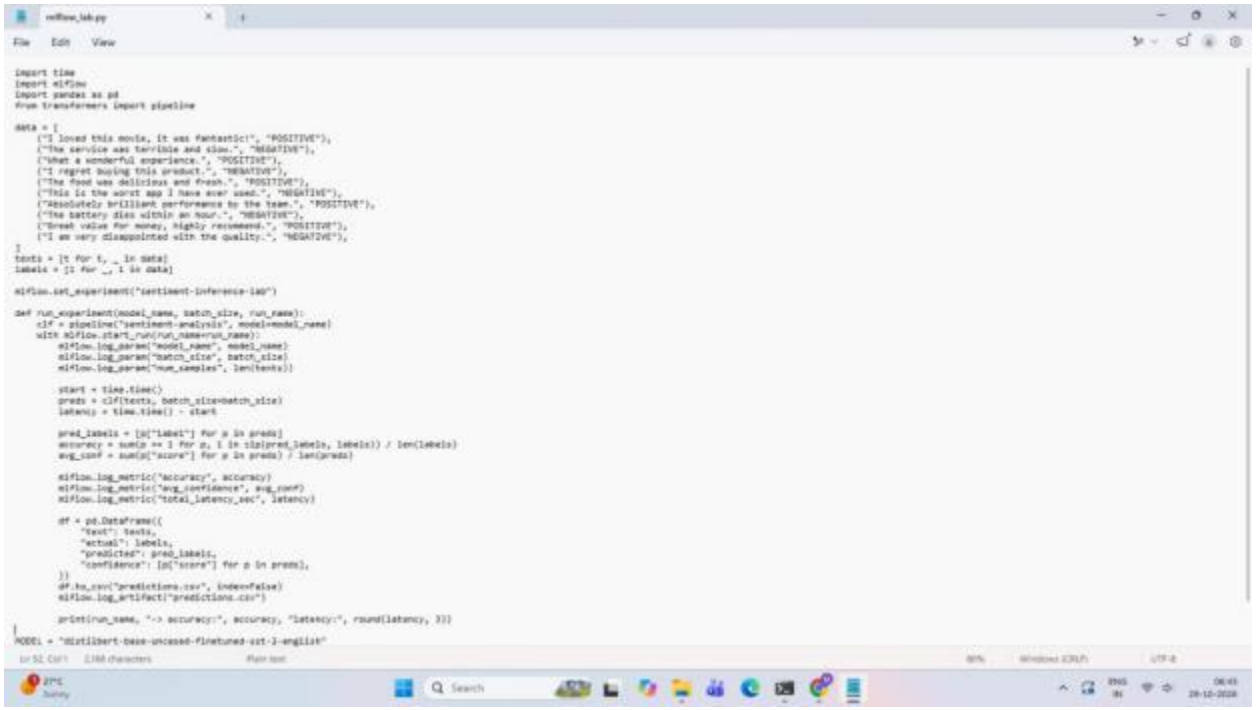

mlflow
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
Metrics (3)
Metric
Value
accuracy
1
avg_confidence
0.9996546826091303
total_latency_sec
0.22608017921447754
Parameters (3)
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
1
num_samples
10
About this run
Created at
10/25/2025, 05:52:43 AM
Created by
GSS
Experiment ID
1
Status
Finished
Run ID
Duration
329ms
Source
mlflow_lab.py
Logged models
Registered prompts
Datasets
None
Tags
Add tags
Registered models
None
Runs
Models
Traces
Model registry
Docs
Settings



Screenshot of the Compare view:



mlflow_lab.py file:



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

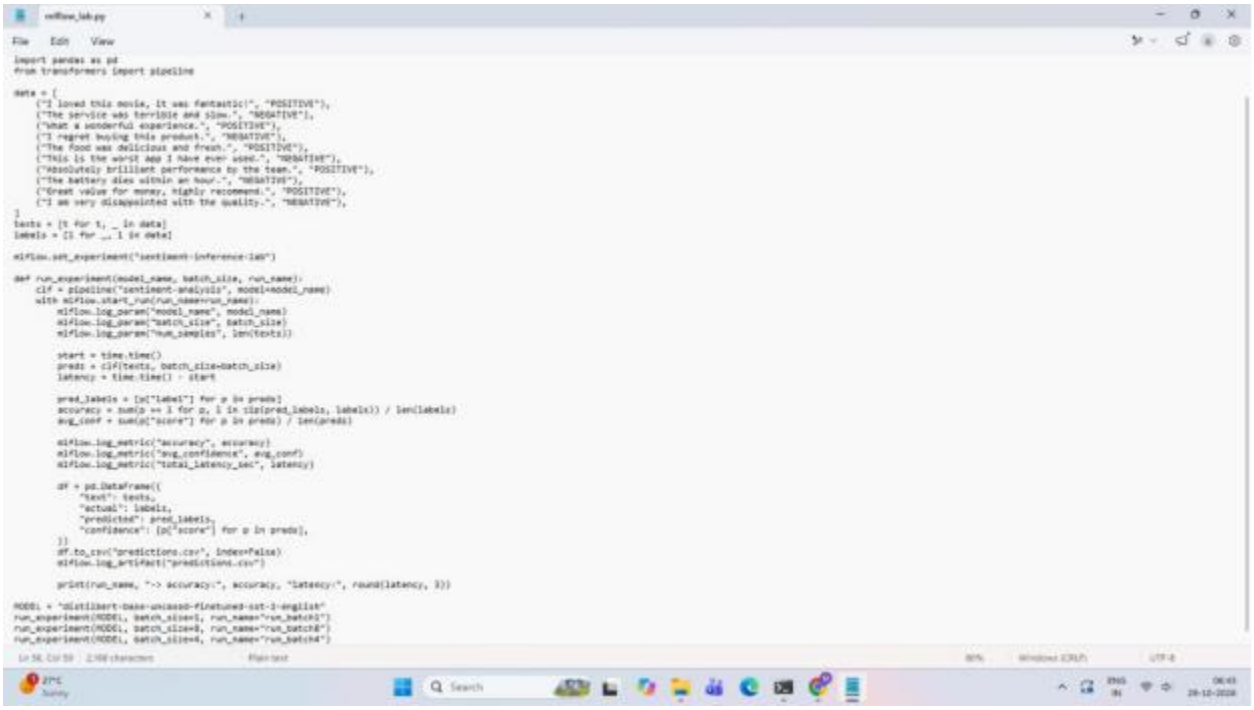
        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
```



```
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

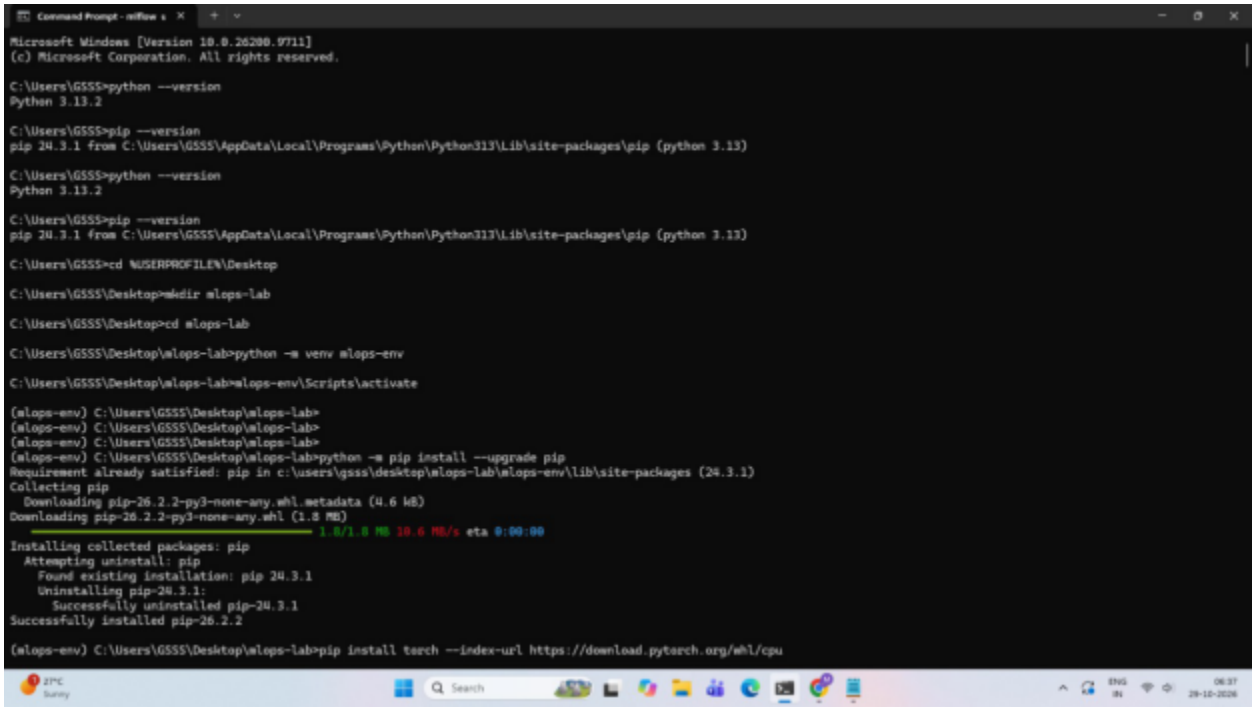
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

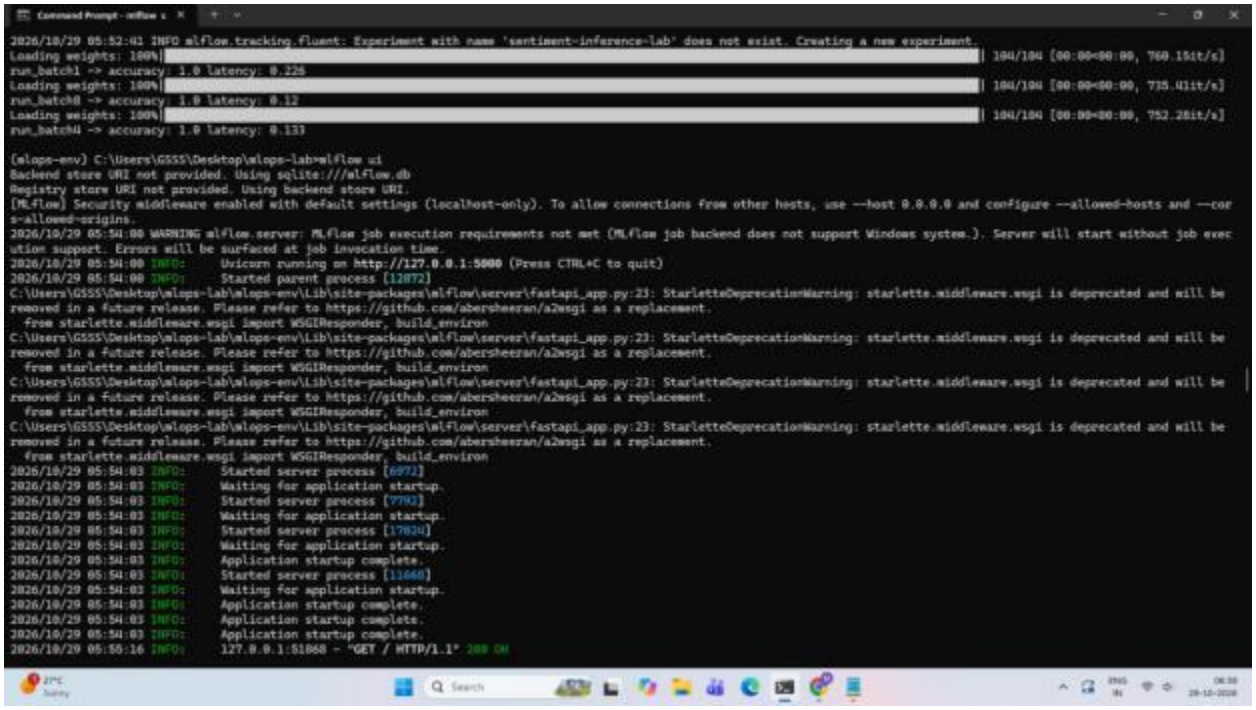
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

Execution:


Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>cd %USERPROFILE%\Desktop
C:\Users\GSSS\Desktop>mkdir mlops-lab
C:\Users\GSSS\Desktop>cd mlops-lab
C:\Users\GSSS\Desktop\mlops-lab>python -m venv mlops-env
C:\Users\GSSS\Desktop\mlops-lab>mlops-env\Scripts\activate
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>python -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\gsss\desktop\mlops-lab\mlops-env\lib\site-packages (24.3.1)
Collecting pip
Downloading pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 10.6 MB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 24.3.1
Uninstalling pip-24.3.1:
Successfully uninstalled pip-24.3.1
Successfully installed pip-26.2.2
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>pip install torch --index-url https://download.pytorch.org/whl/cpu


2026/10/29 05:52:41 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%| 104/104 [00:00<00:00, 760.15it/s]
run_batch1 -> accuracy: 1.0 latency: 0.226
Loading weights: 100%| 104/104 [00:00<00:00, 735.41it/s]
run_batch0 -> accuracy: 1.0 latency: 0.12
Loading weights: 100%| 104/104 [00:00<00:00, 752.28it/s]
run_batch4 -> accuracy: 1.0 latency: 0.133
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 05:54:00 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/29 05:54:00 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 05:54:00 INFO: Started parent process [12872]
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/29 05:54:03 INFO: Started server process [6972]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [7792]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Started server process [17824]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Started server process [11660]
2026/10/29 05:54:03 INFO: Waiting for application startup.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:54:03 INFO: Application startup complete.
2026/10/29 05:55:16 INFO: 127.0.0.1:51868 - "GET / HTTP/1.1" 200 OK